

6.1. $A^T A =$

$$\begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{matrix} 3 \times 2 \\ 2 \times 3 \\ 2 \times 2 \end{matrix}$$

$$\begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix} \quad 2 \times 2$$

Eigen values $\rightarrow \begin{vmatrix} 17-x & 8 \\ 8 & 17-x \end{vmatrix}$

$$\Rightarrow (17-x)^2 - 64 = 0$$

$$\therefore 17-x_1 = 8, \quad 17-x_2 = 8$$

$$\boxed{\lambda_1 \Rightarrow 9, \quad \lambda_2 \Rightarrow 25}$$

2. Singular Values $\rightarrow \sigma_1 = \sqrt{25} = 5 \quad \sigma_2 = \sqrt{9} = 3$

$$\Sigma \Rightarrow \begin{bmatrix} 5 & 0 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}$$

3.

3. $\rightarrow$ Calculating V ,

For $\lambda_1 = 25$

$$(A^T A - 25I) v \Rightarrow 0 \rightarrow \begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} v \Rightarrow 0 \quad \therefore v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_2 = 9 \rightarrow (A^T A - 9I) v \Rightarrow 0 \rightarrow \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} v \Rightarrow 0 \quad \therefore v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$V \Rightarrow \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U \Rightarrow A V \Sigma^{-1} \quad (A, V V^T \Rightarrow I \text{ or } V^{-1} = V^T)$$

$$\therefore \cancel{U} \Rightarrow \cancel{\begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}} \cancel{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}} \cancel{[\quad]}$$

$$u_1 \Rightarrow \frac{1}{\sigma_1} A v_1 = \frac{1}{5} A \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$u_2 \Rightarrow \frac{1}{\sigma_2} A \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$U \Rightarrow \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} \\ 0 & \frac{4}{3\sqrt{2}} \end{bmatrix}$$

Ex 2. $A_1 \Rightarrow \sigma_1 u_1 v_1^T = 5 u_1 v_1^T$ $u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$A_1 \Rightarrow 5 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \dots = \frac{5}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Ex 3 1. $\|A\|_F = \sqrt{\sum a_{ij}^2} = \sqrt{(3)^2 + (2)^2 + (2)^2 + (2)^2 + (3)^2 + (-2)^2}$

$$\Rightarrow \sqrt{34}$$

$$\Rightarrow \sqrt{\sum \sigma_i^2} = \sqrt{5^2 + 3^2} = \sqrt{34}$$

2. $\|A\|_F = \frac{5}{2} \sqrt{(1)^2 + (1)^2 + (1)^2 + (1)^2} = 5$

3. Energy $\propto \|A\|_F^2$

$\therefore \frac{25}{34} \times 100 \rightarrow 73.53\%$ energy is successfully transmitted back.